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The response of *Acropora hyacinthus* and *Montipora tuberculosa* to three different types of colony damage: scraping injury, tissue mortality and breakage

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Abstract

Three types of injuries commonly generated by biotic and abiotic agents on the periphery of coral colonies are tissue mortality, scraping injuries and breakage. To determine if recovery (amount of regeneration) was affected by injury type, colonies of *Acropora hyacinthus* and *Montipora tuberculosa* were inflicted with tissue, scraping and breakage injuries and their recovery monitored over 24 days. The influence of the tissue available for regeneration (i.e. the zone of tissue of bordering or partially injured polyps from which regeneration can occur) and the amount of settlement of algae on recovery was also measured for the different injury types and species. Overall, the amount of regeneration was up to four times greater for a scraping injury than tissue mortality and breakage. Furthermore, the amount of regeneration of the breakage injuries was approximately two-times greater for *M. tuberculosa* than for *A. hyacinthus* colonies. The regeneration rate of all injury types declined over time for both species. Recovery of injuries was influenced by the amount of algae that colonised the lesion and the zone of tissue available for regeneration. In general, the amount of regeneration of injuries was lower for injuries with a small zone of basal tissue available for regeneration, and the amount of regeneration was negatively correlated with the amount of algal settlement. The results of this study are important for understanding the role of partial mortality in the demography of scleractinian corals. Injuries impact on the population dynamics of colonies because regenerative processes require resources

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usually available for other processes (e.g. growth and reproduction), and slow recovery rates potentially increase the chance of whole colony mortality. Published by Elsevier Science B.V.

Keywords: Injury; Regeneration; Scleractinian corals; Type of injury

1. Introduction

Corals provide the basic framework of coral reefs, as well as food and habitats for many associated fauna and flora (e.g. Birkland, 1997). Coral reefs may flourish when disturbances are intermediate in intensity and frequency, and decline if disturbances occur more often and/or with greater magnitude, because coral communities are not given adequate time to recover between impacts (e.g. Connell, 1978, 1996; Hatcher et al., 1989; Chou et al., 1994; Hughes, 1994). The latter scenario has become increasingly common for many reefs under human exploitation (Colloquium and forum on global aspects of coral reefs, Miami, 1993). Unfortunately, while the impact of humans on reefs has been well documented, the underlying mechanisms of effect and their consequences are less understood (Hatcher et al., 1989). The general aim of this chapter is to investigate some of the underlying mechanisms affecting injury and regeneration in scleractinian corals.

Injury in scleractinian corals originates from natural causes (e.g. predation, competition, storm and cyclone damage), and human activities (e.g. anchoring, diving, mining and pollution) and results in the loss of tissue and/or tissue and skeleton from the colony (e.g. reviews by Connell, 1973; Brown and Howard, 1985; Connell, 1996; Brown, 1997). The response of corals to damage is complex and related to both the characteristics of the injury and the life histories of different species (e.g. Bak et al., 1977; Bak and Steward-Van Es, 1980; Meesters et al., 1992, 1997; Meesters and Bak, 1993, 1995; Hall, 1996, 1997; Oren et al., 1997; Marshall, 2000).

The regeneration of injuries is influenced by their position on the colony, and their shape, size and type. The position of an injury on a colony can influence recovery in some colonial organisms such as bryozoans (Palumbi and Jackson, 1982), the gorgonian *Plexaura homomalla* (Wahle, 1983), and scleractinian corals such as *Acropora palmata* (Meesters and Bak, 1995) and *Porites mayeri* (Hall, 1997), but not in others (e.g. *P. astreoides*, Meesters et al., 1992; *A. robusta*, *A. hyacinthus*, *A. palifera*, *Pocillopora damicornis* and *P. lichen*, Hall, 1997). Variations in the recovery of injuries located at different positions within a colony have been associated with the senescence of polyps (and zooids) along a proximal to distal gradient (e.g. Palumbi and Jackson, 1982; Meesters and Bak, 1995), reduced light levels as a result of shading (Meesters et al., 1992), different polyp densities at the base and top of massive colonies (Darke, 1991; Hall, 1997) and substratum interactions (Jackson, 1979).

Recovery of injuries is size-specific with regeneration being higher for small (1 cm^2) injuries as compared with larger (5 cm^2) injuries (Bak et al., 1977; Bak and Steward-Van Es, 1980). However, the shape of an injury is as important as its size since recovery is influenced by both its area and the amount of healthy tissue bordering its perimeter

(Meesters et al., 1997; Oren et al., 1997). Meesters et al. (1997) have suggested that resources for recovery come from those polyps bordering the lesion, while Oren et al. (1997) have proposed that these resources may be translocated from both neighbouring and more distant polyps, depending on the length of the injury. These conclusions have been drawn from studies that have examined only one specific injury type and where regeneration occurs at the edge of the injury. Meesters et al. (1997) removed the tissue and skeleton from a colony and Oren et al. (1997) removed only the tissue, leaving the skeleton intact. The amount of damage incurred, and the area of tissue from which regeneration can occur, however, are dependent on the types of injury inflicted.

Coral injuries can be divided into four major categories: partial tissue loss, tissue mortality, superficial tissue and skeleton loss (e.g. scraping injuries), and substantial tissue and skeleton loss (e.g. breakage) (see Hall, 1997). The recovery of these different types of injuries varies enormously. Bak et al. (1977) showed that the recovery of tissue injuries was faster than scraping injuries in *Agaricia agaricites* while the opposite pattern was true for *Montastrea annularis*. Similarly, recovery of scraping injuries was faster than the recovery of tissue injuries for *P. australiensis* and *A. cytherea* (Hall, 1997). In a comparison of breakage versus tissue injuries, the regrowth of a branch was 10 times slower than the recovery of a tissue denuded branch (Hall, 1997). Recovery of different injury types appears linked to the amount of tissue and skeleton that must be replaced, as well as the ability to out compete organisms that have settled onto the injured area.

The morphology of a colony often dictates its susceptibility to different types of injury, and influences the subsequent size of any injury. For example, branching species are more prone to storm and diver damage than massive species (Woodley et al., 1981; Hughes, 1989; Glynn, 1990; Chadwick-Furman, 1995; Marshall, 2000), and massive and encrusting species are more vulnerable to overgrowth by algae (Hughes, 1989). For branching species, differences in branch thickness can result in different sized lesion if they are broken off. This has been demonstrated for *A. gemmifera* and *A. millepora* where the resulting injury to the colony through breakage was three times larger for the thicker-branched *A. gemmifera* (Hall, 1997).

If the injured area does not regenerate within a couple of days, algae and other organisms will settle onto the injured area (e.g. Bak et al., 1977). These organisms may then compete with the coral for food and space. As a consequence, recovery may be slowed since resources available for regeneration may be reallocated to competition. In the long-term, colony survivorship may be affected if (1) the colonising organism is a superior competitor, (2) the algal patch is used as an entry site for boring organisms that disrupt the integrity of the colony, or (3) the algae provides an infection site for diseases (Bak et al., 1977; Palumbi and Jackson, 1982; Wahle, 1983; Hughes and Jackson, 1985; Babcock, 1991).

The ability of many corals to recover from injuries, and the effect of position, size, and shape of injuries on the recovery process have been well documented (e.g. Stephenson and Stephenson, 1933; Bak et al., 1977; Bak and Steward-Van Es, 1980; Bak, 1983; Wahle, 1983; Rinkevich and Loya, 1989; Meesters et al., 1992, 1994; Meesters and Bak, 1993, 1995; Hall, 1997; Oren et al., 1997; Marshall, 2000). In contrast, the influence of injury type on recovery has largely been ignored (but see Bak

et al., 1977; Bak and Steward-Van Es, 1980). The aim of this paper is to investigate how recovery of damage is influenced by injury type for *A. hyacinthus* and *M. tuberculosa*. Specifically, I will examine if (1) the amount of colonisation of the injured site by algae, (2) the dimensions of the tissue from which regeneration can occur, and (3) regeneration rates, vary among colonies with tissue mortality, scraping or breakage injuries for the two species of corals.

2. Materials

To investigate the response of corals to scraping injuries, tissue mortality and breakage, I injured colonies of two species and monitored the subsequent recovery process. The species were *A. hyacinthus* and *M. tuberculosa*, chosen because of their susceptibility to the injury types under investigation, their contrasting morphologies, and abundance on the reef-crest at Mermaid Cove, Lizard Island, Australia, (14°40′,145°28′) where the study was conducted. *A. hyacinthus* is a fast-growing, relatively short-lived, tabular coral consisting of a wide flat table of fused branches from which fine upwardly projecting branchlets protrude. *M. tuberculosa* is also a relatively short-lived, fast-growing coral but has a plate-like, foliaceous morphology.

Different techniques were used to generate the three different types of injuries on colonies. For tissue injuries, all tissue was removed from a section of the colony by blowing it away from the skeleton with compressed air. This air was discharged from the nozzle of an air-gun at a constant pressure of 60 psi. A template cut from water-proof paper was used to protect surrounding tissue from damage and to standardise the injury size. Scraping injuries were inflicted by lightly dragging a fine chisel over the surface of colonies to remove a superficial layer of tissue and skeleton (1–2 mm), including structures protruding above the coenosteum. Breakage was simulated by carefully chiselling away a small section of the colony. The size, shape and position of the injury on the colony (Fig. 1a), and colony size, were standardised across all three injury regimes since these characteristics can affect recovery (Loya, 1976; Bak et al., 1977; Meesters and Bak, 1995; Meesters et al., 1997; Oren et al., 1997). All injuries were relatively small, having a mean projected area of $3.3 \pm$ SE 0.13 cm^2. Injuries were inflicted in situ on the edge of colonies and only large colonies (diameter > 75 cm) were used. A total of 60 colonies was injured in this experiment (10 per injury type and species). The positions of all colonies were marked with horticultural tags attached to the substratum close to the colony to ensure relocation at subsequent censuses.

2.1. Amount of regeneration

To determine the amount of regeneration of the injuries (recovery) over time, the injured region of each colony was monitored photographically with a Nikonos V underwater camera. Close-up photographs, using a 35-mm close-up attachment, were taken at the time of the injury, and 12 (census 1) and 24 (census 2) days after the injuries were inflicted. A ruler attached to the close-up frame was used as a size

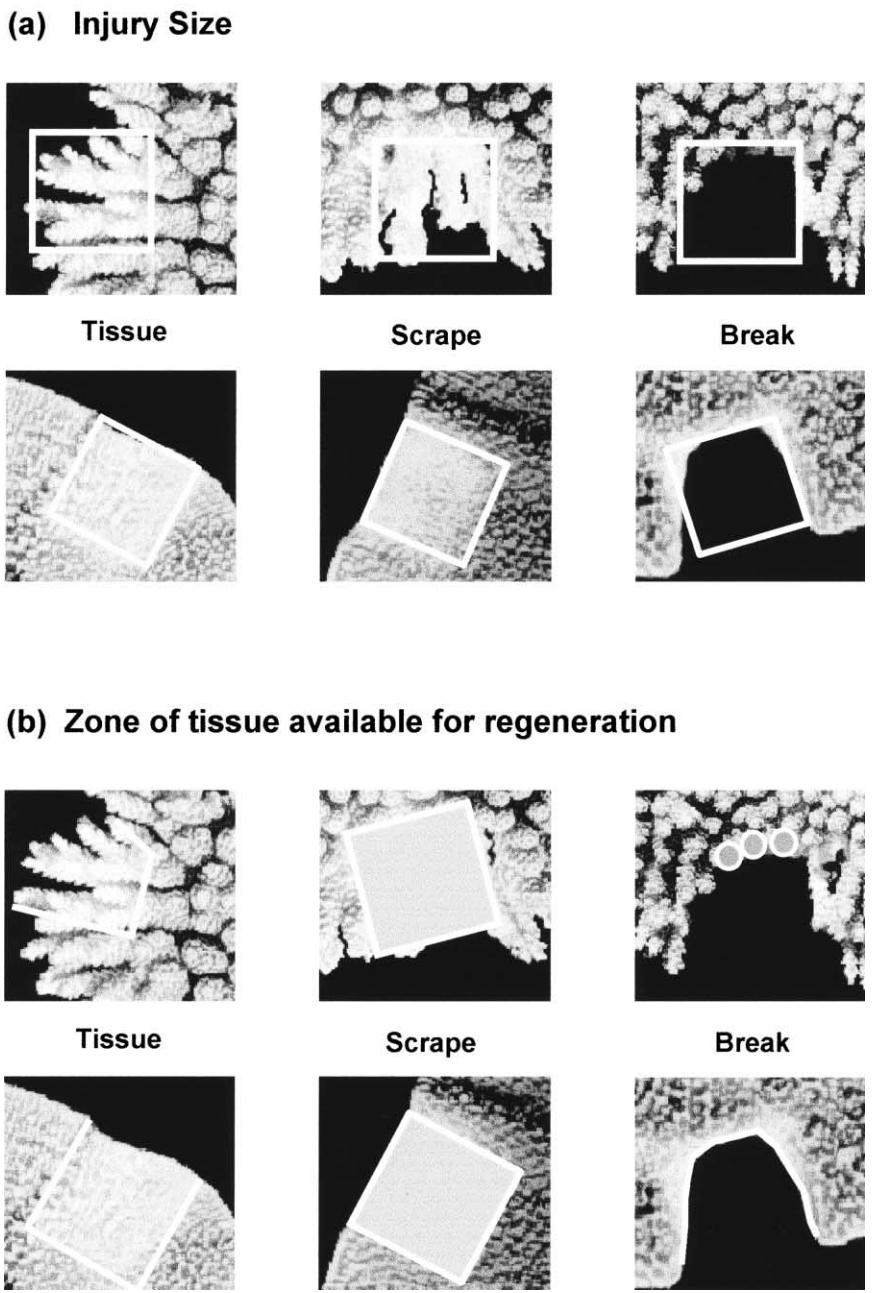

Fig. 1. Photographs of the different injury types for *A. hyacinthus* (top row) and *M. tuberculosa* (bottom row). (a) The size of the injury is outlined in white, and (b) zone of tissue available for regeneration is outlined in white with the respective areas shaded in grey. For scale, the side of the box = 1.8 cm.

reference. The amount of regeneration that had taken place over time was quantified from the photographs by projecting the image at its actual size onto a paper screen and tracing the damaged and recovered areas. Both areas were then digitised on a Hi-Pad-Plus Digitiser used in association with the computer package Sigma-Scan. The amount of regeneration that occurred over time has been presented graphically as a percentage of the original injury size for each injury type.

2.2. Amount of algal settlement

To examine the effect of colonisers on the recovery of the three different injury types, close-up photographs of the amount of algae (the initial coloniser of damaged areas) settling onto the injured areas were taken 12 and 24 days after injury. The amount of algae present on injuries over time was then determined by tracing and then digitising the areas colonised by algae (as described previously for regeneration). The amount of algae present has been presented graphically as a percentage of the original injury area for tissue and scraping injury, and as a percentage of the viable area of regeneration for breakage injuries since viable area of regeneration represents the area of damage on the colony and thus the potential site for colonisation.

2.3. Zone of tissue available for regeneration

The modes of regeneration for the three different injury types were examined by observing the patterns of regeneration from consecutive photographs of injuries. The influence of bordering or partially damaged polyps within the injury site from which regeneration can take place (zone of tissue area available for regeneration) on the recovery of different types of injuries was then quantified by measuring the dimensions of the tissue involved in regeneration (see Hall, 1997) with a set of vernier callipers. For tissue injuries, regeneration occurs from the margins of the injury so that the perimeter of the injury (minus the outer edge) was measured (Fig. 1). For breakage injuries, regeneration was activated from the area of tissue that remains after the break. The shape of this area varied with species. To obtain the dimensions of the tissue available for regeneration for breakage injuries for each species, the perimeter and depth of the bordering tissue was measured for *M. tuberculosa*, and the diameter of the base of branchlets was measured for *A. hyacinthus* (Fig. 1b). For scraping injuries, the dimensions of the injury were measured since regeneration occurs from the underlying partially damaged polyps (see Hall, 1997). Geometric formulas for perimeters, rectangles and circles were used to calculate the dimensions of the tissue from which regeneration can take place.

2.4. Analysis of data

Differences between species and injury types in the amount of regeneration and algae present over 12 and 24 days were tested using a repeated measures ANOVA. This analysis tests for between subject effects (i.e. differences between species and injury types for the two variables) and within subject effects (differences in the variables

between census one and two for species and injury types). Variables were log transformed to meet the assumptions of the analysis. For each injury type, differences in the amount of tissue available for regeneration between species were tested with a *t*-test. (A separate analysis was conducted for each injury types since the mode of regeneration differed among injury types, see Hall, 1997.) This data set was log transformed to meet the assumptions of the *t*-test. When differences in amount of tissue available for regeneration between species were significant, a repeated analysis of covariance (ANCOVA) was subsequently carried out on the amount of regeneration over 12 and 24 days, with amount of tissue available for regeneration as the co-variate. This analysis factors out the effect of amount of tissue available for regeneration on recovery. A Pearson's Correlation Analysis was used to determine if there was any association between the amount of regeneration of injuries and the amount of algal settlement on injuries. The data were log transformed to meet the assumptions of the analysis. Tukey's Test was used as a post-hoc test to distinguish between significant differences in main effects and interactions identified by the ANOVAs.

3. Results

Striking differences were detected in the interaction between species and injury type in the regeneration of injuries (Table 1, Fig. 2). This pattern was influenced by the amount of tissue contributing to recovery and the level of colonisation of algae onto lesions (Tables 2, 3 and 4; Figs. 3 and 4). Furthermore, the pattern of recovery and colonisation changed over the two census times, especially among injury types (Tables 1 and 2; Figs. 2 and 3).

Table 1
Summary of the repeated measures ANOVA testing for differences in regeneration between species and injury types (Between Subject Effects) and how these patterns change between censuses (Within Subject Effects)

Source of variation	*df*	Type III SS	Mean square	*F*-ratio	Probability
Between Subject Effects					
Species	1	0.041	0.041	2.58	0.115
Injury Type	2	2.177	1.088	68.86	0.000
Species * Injury Type	2	0.132	0.066	4.19	0.021
Error	47	0.743	0.016		
Tukey's Test	*A. hyacinthus* scrape, *M. tuberculosa* scrape, *M. tuberculosa* break, *M. tuberculosa* tissue, *A. hyacinthus* tissue, *A. hyacinthus* break				
Within Subject Effects					
Census	1	0.091	0.091	41.17	0.000
Census * Species	1	0.005	0.005	2.31	0.104
Census * Injury Type	2	0.032	0.016	7.31	0.002
Census * Species * Injury	2	0.012	0.006	2.69	0.079
Error	47	0.104	0.002		

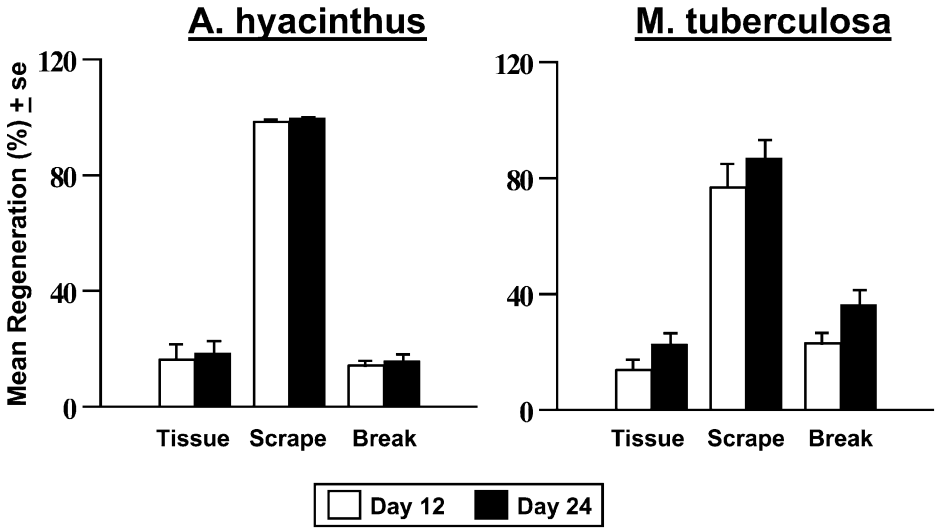

Fig. 2. Mean regeneration of injuries (expressed as a percentage of the original injury size) for injury type and species after 12 and 24 days. (Error bars represent 1 SE).

3.1. Amount of regeneration

Rates of recovery varied markedly between injury types, and also between species for breakage injuries (Between Subject Effects, Table 1, Fig. 2, Tukey's Test) over 24 days. In contrast, there was no difference in recovery for scraping and tissue injures for *A. hyacinthus* and *M. tuberculosa* (Fig. 2, Tukey's Test). The amount of regeneration of scraping injuries was high during the course of this study (> 75% of the original injury, Fig. 2). This was in marked contrast to the recovery of tissue and breakage injuries where less than 40% of the injured area was regenerated (Fig. 2). However, although the regrowth of lost polyps was slow for breakage injuries, most of the section of the colony that was damaged as a result of breakage was rapidly sealed over within 12 days. For breakage injuries, the amount of recovery was approximately twice as high for *M. tuberculosa* as *A. hyacinthus* (Fig. 2). The overall amount of recovery of all injuries types was much greater in the first 12 days as compared to the second 12 days (Within Subject Effects for census, Table 1, Fig. 2). This pattern of regeneration among censuses was consistent for species but not injury types (Within Subject Effects, Table 1). For injury types, the amount of recovery between censuses was approximately 1.2–1.5 times greater for tissue injuries and breakage as compared to scraping injuries (Fig. 2).

3.2. Amount of algal settlement

Large differences were detected in the amount of algae that colonised the injured area in 24 days (Between subject effects, Table 2, Fig. 3). Algal cover on tissue injuries (bare

Table 2
A summary of the repeated measures ANOVA testing for differences in the amount of algal colonisation between species and injury types (Between Subject Effects) and how these patterns change after 12 and 24 days (Within Subject Effects)

Source of variation	*df*	Type III SS	Mean square	*F*-ratio	Probability
Between Subject Effects					
Species	1	0.026	0.026	2.60	0.114
Injury Type	2	7.724	3.862	392.59	0.000
Species * Injury Type	2	0.057	0.028	2.87	0.066
Error	47	0.462	0.010		
Within Subject Effects					
Census	1	0.027	0.027	13.75	0.001
Census * Species	1	0.003	0.003	1.56	0.218
Census * Injury Type	2	0.013	0.006	3.29	0.046
Census * Species * Injury	2	0.005	0.003	1.32	0.277
Error	47	0.094	0.002		

skeleton) for both species was greater than 70% after 12 and 24 days, and less than 1% for breakage injuries after 12 and 24 days (Table 2, Fig. 3).

For colonies with scraping injuries, algal cover was low for *A. hyacinthus* and intermediate for *M. tuberculosa* ($< 2\%$ and $< 20\%$, respectively) after 12 and 24 days. The amount of algae present at census 2 was much less than the amount present after census 1 (Census, Within subject effects, Table 2). There was a consistent pattern in change of algal cover over time between species but not injury type (Within subject

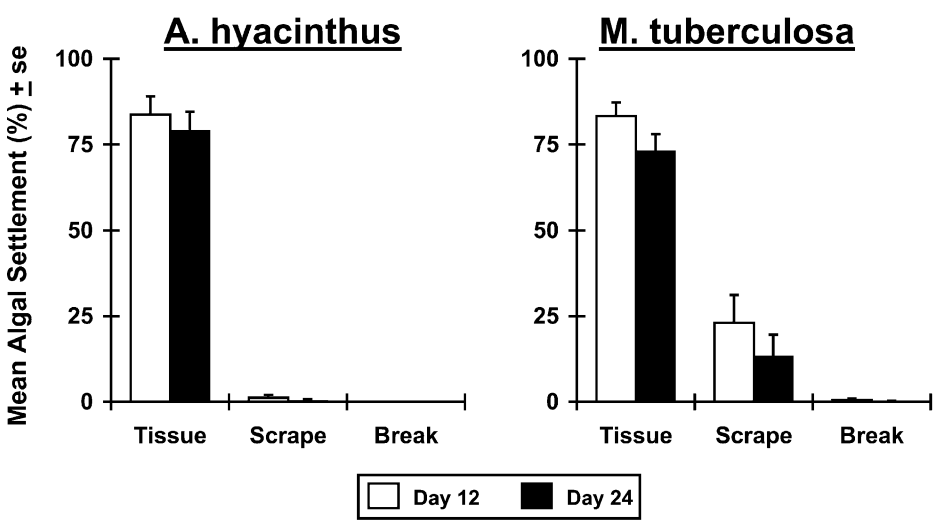

Fig. 3. Mean algal settlement expressed as a percentage of the original injury size for injury types and species after 12 and 24 days. (Error bars represent 1 SE).

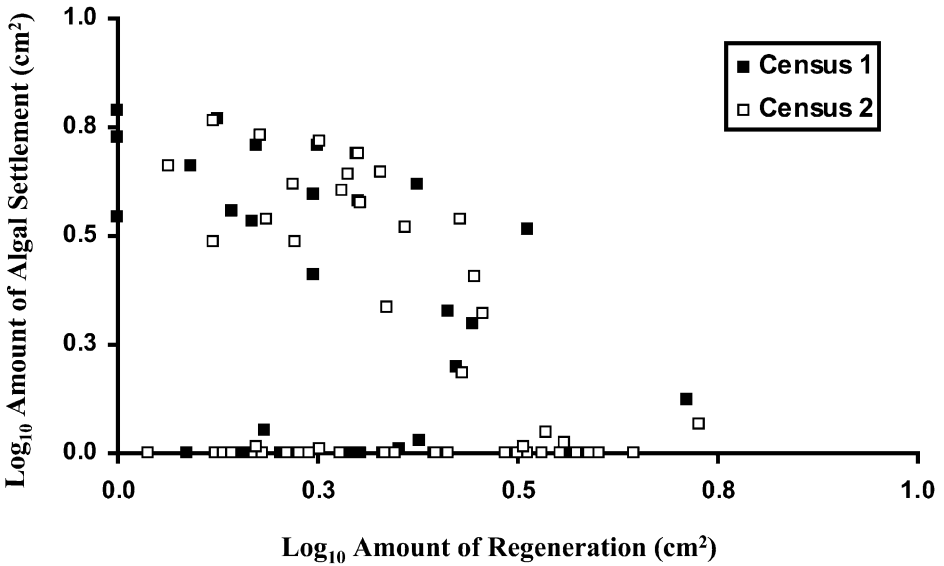

Fig. 4. A scatter plot showing the correlation between amount of regeneration (cm^2) and amount of algal settlement (cm^2) over 12 and 24 days.

effects, Table 2). Algal cover declined by 8–10% for tissue injuries, 50% for scraping injuries and more than 50% for breakage between census 1 and census 2 (Fig. 3). There was a negative relationship between the amount of regeneration and amount of algal cover (Pearson's Correlation Coefficient = −0.441 ($P = 0.001$) and −0.348 ($P = 0.011$) for day 12 and 24, respectively (Fig. 4).

3.3. Zone of tissue available for regeneration

The amount of tissue available for regeneration reflects differences in the modes of regeneration (Table 3). For scraping injuries, the zone of tissue area available for regeneration was equal to the original size of the injury since the partially damaged polyps within the injury contributed to the regeneration process (Table 3, Fig. 1). In

Table 3
The zone of tissue available for regeneration ± 1 SE for each injury type and species, and the results of the *t*-test analysing differences between species in the amount of tissue available for regeneration for each injury type

Injury type	Zone of regeneration	Species		*T*-test
		A. hyacinthus	M. tuberculosa	Statistic (*df*), *P*
Tissue	Perimeter of injury (cm)	6.00 ± 0.21	6.19 ± 0.54	0.72 (18), 0.481
Scrape	Area of injury (cm^2)	3.23 ± 0.10	1.00 ± 0.00	0.47 (17), 0.642
Break	Area at base of break (cm^2)	0.45 ± 0.08	1.66 ± 0.26	16.13 (17), 0.000

Table 4
A summary of the repeated measures analysis of covariance testing for differences in regeneration between *A. hyacinthus* and *M. tuberculosa* for breakage injuries (Between Subject Effects), and how these patterns change between censuses (Within Subject Effects) after the effect of amount of tissue available for regeneration (ATR) has been adjusted for

Source of variation	*df*	Type III SS	Mean square	*F*-ratio	Probability
Between Subject Effects					
Species	1	0.016	0.016	1.67	0.214
ATR	1	0.049	0.049	5.18	0.037
Error	16	0.152	0.010		
Within Subject Effects					
Census	1	0.016	0.016	18.34	0.001
Census * Species	1	0.001	0.001	0.53	0.479
Census * ATR	1	0.000	0.000	0.11	0.743
Error	16	0.014	0.0019		

contrast to scraping injuries, regeneration of tissue and breakage injuries only occurred at the margins of the lesions since polyps within the lesions were either dead or had been removed (Fig. 1). For tissue injuries, the tissue at the perimeter of the injury formed a smooth layer of tissue and skeleton which gradually grew inwards, thus enveloping the injured site. Polyps were then produced along the margin of the band that was closest to the uninjured tissue. For breakage injuries, recovery was facilitated by the extension of the skeleton via the growth of new polyps.

The amount of tissue available for regeneration was consistent among species for tissue mortality and scraping injuries but varied markedly between species for breakage injuries (Table 3). For breakage injuries, the tissue area contributing to regeneration was approximately three times higher for *M. tuberculosa* than *A. hyacinthus* (Table 3). This difference occurred because *M. tuberculosa* has a plate morphology and the margin for regeneration is a continuous U-shape while *A. hyacinthus* is made up of discrete branchlets, and regeneration only occurs from the base of these individual branchlets (Fig. 1). This interspecific difference in amount of tissue available for regeneration clearly had an effect on the amount of regeneration over 24 days since the significant difference in regeneration is negligible once the effect of amount of tissue available for regeneration has been adjusted for by ANCOVA (Between subject effects, species, Table 4). Thus, this result suggests that the recovery of breakage injuries is similar among species providing the area available for regeneration is the same.

4. Discussion

The capacity to recover quickly from damage was variable and dependent on the interaction between species and injury type. Recovery was influenced by the amount of tissue available for regeneration and the proportion of the injury that was settled by algae. Interspecific differences in the amount of regeneration of different injury types

have consequences for the dynamics of coral populations under varying injury regimes since injury can affect fitness.

The amount of tissue available for regeneration is dependent on the type and level of damage inflicted and interspecific variations in colony morphology. The amount of regeneration of lesions differed among injury types for both species and was three and four times higher for scraping injuries as compared with tissue mortality and breakage (Fig. 2). The amount of tissue available for regeneration was influenced by injury type since different injury types cause varying amounts of damage to colonies, and their subsequent modes of regeneration vary. The damage caused to colonies by scraping was superficial so that the underlying polyps were only partially damaged and contributed to the regeneration process (Hall, 1997). In contrast, the recovery of tissue and breakage injuries only occurred at the margins of the injury via the growth of new polyps. This growth was initiated by the surrounding healthy polyps although it is possible that some of the resources necessary for regenerative processes may have come from polyps further away through translocation (see Oren et al., 1997; Hall, 1998). The differences detected in recovery between injury types suggests that it is energetically more costly to produce new polyps than to repair partially damaged ones.

The interaction between species and injury type was due to interspecific differences in the amount of regeneration of breakage injuries where the amount of regeneration was three times larger for *M. tuberculosa* than for *A. hyacinthus* (Fig. 2). This pattern was solely due to differences in the amount of tissue available for regeneration since (1) differences in the amount of regeneration disappear when the zone of tissue available for regeneration is accounted for by ANCOVA (Table 4), and (2) the amount of algae settling on this type of injury was negligible (Fig. 3). This result highlights how the amount of damage incurred, and the subsequent amount of tissue available for regeneration is dependent on morphology. *M. tuberculosa* has a plate morphology while *A. hyacinthus* has a tabular morphology so that the removal of a square section from the outer edge of colonies of these two species resulted in two completely different areas from which regeneration could occur. For *M. tuberculosa*, regeneration of new tissue and skeleton occurred from the U-shaped edge created by breakage while for *A. hyacinthus* colonies, regeneration could only take place from the base of branchlets that had been removed (Fig. 1, Hall, 1997). As a consequence, for breakage events, the symmetry of the plate coral is restored a lot faster than that of the tabular coral. Maintenance of the symmetry of a colony in non-encrusting corals is important for such processes as defence, feeding, and mechanical stability (Jackson, 1979; Hughes and Jackson, 1985).

The settlement of algae onto the injured area appeared to influence recovery because there was a negative correlation between the amount of regeneration and algal cover over 12 and 24 days (Fig. 4). The overall pattern of a reduction in regeneration with an increase in algal settlement suggests a trade-off in available resources between regeneration and competition. Differences in the amount of algae that settled onto injury sites may be linked to the production of mucus since mucus appears to act as a deterrent to the colonisation of damaged areas on corals by other organisms (Benson et al., 1978; Rutzler et al., 1983). Settlement by algae was relatively low for scraping and breakage injuries since tissue that had been partially damaged retained its ability to produce

mucus. Conversely, the removal of mucus producing tissue resulted in a high settlement of algae as observed for the tissue injuries. Oren et al. (1997) proposed that the tentacles from surrounding healthy polyps may offer protection from the settlement of algae and sediment for small injuries with high perimeter to surface area ratios. Consequently, tentacles from healthy polyps may have protected breakage injuries from algal settlement due to a high perimeter to surface area ratio of the injury. However, this ratio would have been too low for this process to be effective for scraping injuries.

Recovery of injuries declines over time (Bak, 1983; Meesters et al., 1992, 1994, Fig. 2). This decline in regeneration may result from the exhaustion of a finite supply of resources available for regeneration (Meesters et al., 1994; Rinkevich, 1996), and/or an increasing loss over time of these available resources to competitive interactions with algae. Algae may further impede the process of regeneration over time because it steadily increases in biomass and acts as a trap for sediment. Consequently, the presence of a solid mat of algae and sediment may present an obstruction difficult for the coral to overgrow.

In general, the results of this study suggest that tissue injuries are more detrimental to the long-term survival of colonies than scraping and breakage injuries because their recovery was slower. Recovery of tissue injuries was hindered by both their greater susceptibility to colonisation by algae and a smaller zone of tissue available for regeneration. As a result, the injured area of the colony is exposed to colonisers for longer periods of time, thus increasing the probability of settlement by a superior competitor, infection by pathogens, or invasion by bioeroders (e.g. Bak et al., 1977). All these factors increase the risk of total mortality in colonies (Connell, 1973).

Breakage injuries may also impact on the population dynamics of some coral species since slow recovery rates will impede on the speed at which the symmetry of a colony is restored. Colony symmetry is important for defensive and feeding processes, and mechanical stability (Jackson, 1979; Hughes and Jackson, 1985). However, while breakage may be costly to individual colonies within a population, its effects on the genet population may be favourable in some species since breakage can potentially promote asexual reproduction through fragmentation. Nevertheless, the cost of breakage, if it occurs to a significant proportion of colonies within a genet population, is very likely to outweigh any benefits of fragmentation due to decreased absolute growth rates and decreased reproductive output.

In contrast to tissue and breakage injuries, scraping injuries seem to have the least impact on the population dynamics of corals. The advantage of faster regeneration rates is that the functional use of polyps (photosynthesis, defence, reproduction, feeding and storage) is restored more quickly (Jackson, 1979; Sebens, 1987; Hughes and Jackson, 1985), and the risk of invasion by competitors, pathogens and bioeroders is minimised (Jackson and Palumbi, 1979; Wahle, 1983; Hughes and Jackson, 1985; Sebens, 1987).

In conclusion, the capacity to recover quickly from damage differed among species and injury types and was influenced by the amount of tissue available for regeneration and the proportion of the injury that was settled by algae. Generally, the amount of regeneration of injuries was lower for injuries with a small amount of tissue available for regeneration and/or high levels of algal settlement. The findings of this study are important for understanding the population dynamics of corals since slow recovery rates

increase the chance of whole colony mortality and regeneration processes deplete resources normally available for growth and reproduction.

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